

A Training-Free Long-Video Understanding Agent with Structured Memory and Temporal Re-observation

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Abstract

Long-video question answering is difficult because the amount of visual information often exceeds the effective visual context that a multimodal model can process in one pass. Uniform frame sampling is a simple baseline, but it may waste visual budget on redundant frames while missing short-lived evidence. This report presents a training-free long-video understanding agent built around a frozen Qwen3-VL base model. The system uses temporal re-observation as a multimodal tool, stores reusable 60-second structured segment memory, retrieves question-relevant segments at query time, expands neighboring temporal context, and performs final answering from local visual observations with the selected memory records retained as auxiliary guidance. On a 30-question subset sampled from three LVBench long videos, uniform 32-frame and 64-frame baselines both achieve 33.33% accuracy. A multi-window agent without memory improves to 43.33%, while the final Structured Memory + Neighbor Expansion agent reaches 53.33%. Retrieval analysis further shows that neighbor expansion increases temporal evidence coverage from 66.67% to 83.33%. These results suggest that effective long-video QA depends less on uniformly increasing frame count and more on how visual budget, memory, temporal retrieval, and local perception are organized. The code is available at <https://anonymous.4open.science/Video-Understanding-Agent>

1 Introduction

Long-video understanding requires a model to answer questions whose evidence may appear anywhere in a video lasting tens of minutes or even longer. A direct solution is to uniformly sample frames from the whole video and feed them to a multimodal model. However, this strategy has two limitations. First, many sampled frames can be redundant because nearby frames or visually similar scenes may repeat. Second, short-lived evidence such as a piece of visible text, a

brief action, or a counting cue can easily fall between sampled frames.

This project studies a training-free alternative. Instead of retraining the base model, I design an agentic harness around the same Qwen3-VL model [1]. The central idea is to treat long-video QA as a process of information acquisition: the system should decide which temporal evidence is likely to matter, call tools to observe those regions more carefully, store reusable memory for the same long video, and separate long-video perception from final local visual reasoning when useful.

The final system combines three components. **Temporal re-observation tool** samples local frames from selected windows rather than uniformly increasing global frame density. **Structured segment memory** stores reusable video-level information at 60-second granularity. **Neighbor expansion** step extends retrieved memory segments to adjacent time ranges before local visual QA, reducing evidence loss caused by segment boundaries.

The main findings are:

- Uniformly increasing frames from 32 to 64 does not improve accuracy on a 30-question evaluation subset (LVBench-30).
- Multi-window temporal re-observation improves accuracy under the same 32-frame visual budget.
- Structured memory and neighbor expansion further improve both accuracy and temporal evidence coverage.
- Remaining failures shift from coarse temporal localization to fine-grained local evidence understanding, especially OCR, counting, and action-chain reasoning.

2 System Overview

Figure 1 contrasts conventional uniform sampling with the proposed structured-memory temporal re-observation agent using a real LVBench example. The upper frame ribbon visualizes the full video rather

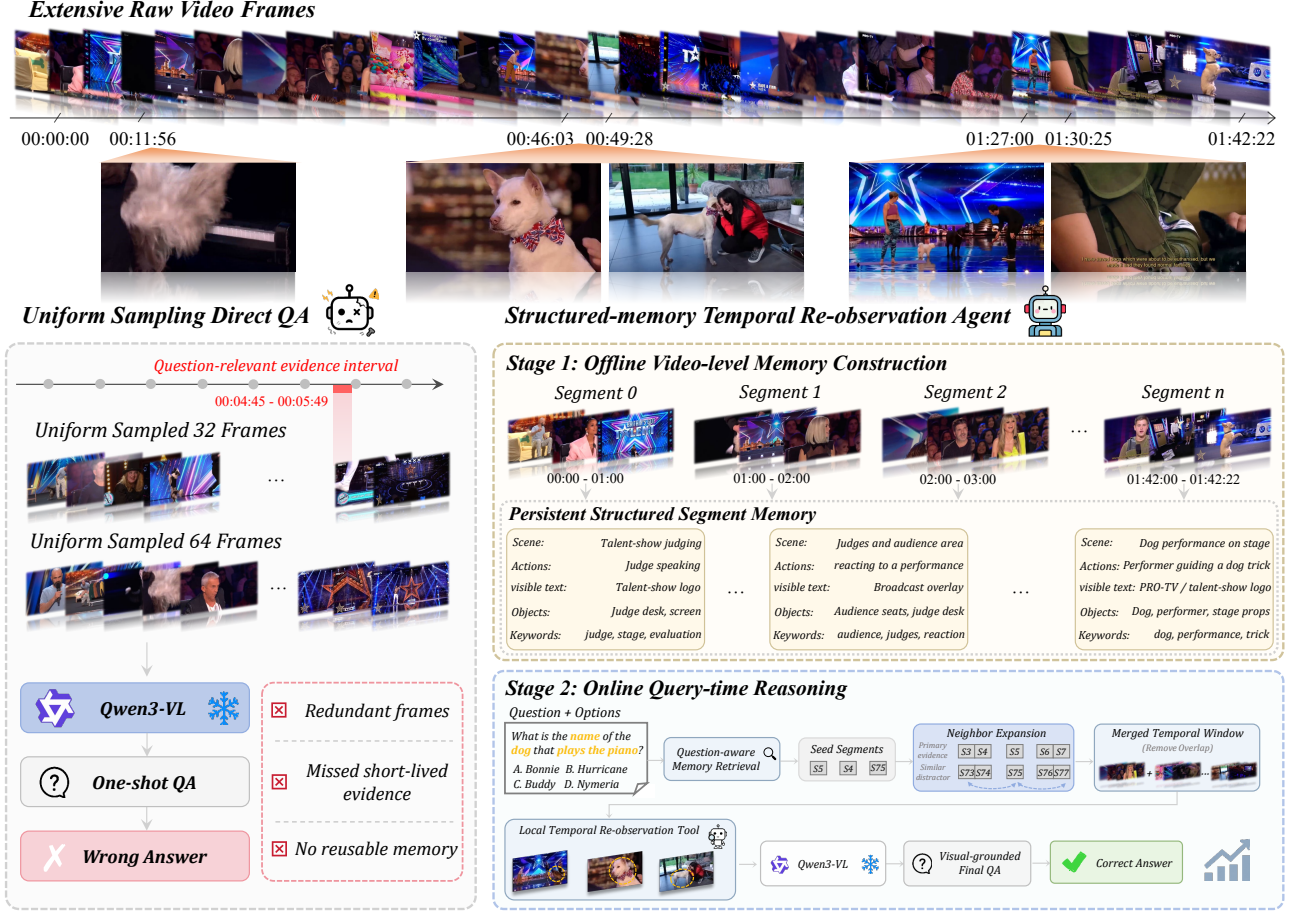


Figure 1. Overall framework with real LVBench video frames. The upper strip presents representative frames sampled across the long video. The left panel illustrates question-agnostic uniform sampling with U32/U64 followed by one-shot QA. The right panel presents the proposed pipeline, including offline structured memory construction, question-aware retrieval, neighbor expansion, local temporal re-observation, and visual-grounded final QA. The structured memory identifies where relevant evidence may appear, while the re-observed original frames provide the visual basis for the final answer.

than the actual model input, illustrating that long videos contain diverse scenes while question-relevant evidence may occupy only a brief interval. The left branch performs question-agnostic U32/U64 sampling followed by one-shot QA, whereas the right branch uses a reusable structured memory to guide local visual re-observation.

The proposed agent operates in two stages. During offline processing, the video is divided into fixed-duration segments and converted into a persistent structured memory bank. During online reasoning, the question and candidate answers retrieve seed segments, neighboring context is expanded and merged, and the original video is re-observed within the resulting temporal windows. The final prediction is generated from the local visual observations together with the selected memory records, while the prompt explicitly treats the original frames as the primary evidence and the memory as auxiliary guidance.

In the illustrated example, the retrieved segments S_5 , S_4 , and S_{75} produce two expanded temporal groups, S_2 – S_7 and S_{73} – S_{77} . The first group overlaps the annotated evidence interval from 00:04:45 to 00:05:49, enabling local visual re-observation to support the correct answer, Bonnie.

3 Method

This section formalizes the observation, memory, retrieval, and answer-prediction mechanisms summarized in Figure 1.

3.1 Uniform Sampling Baseline

For a video of duration T , the uniform-sampling baseline selects N timestamps distributed across the complete video:

$$t_i = \frac{i + 0.5}{N}T, \quad i = 0, \dots, N - 1. \quad (1)$$

The sampled frames are provided to the frozen vision-language model together with the question and candidate answers for one-shot prediction. Two settings, $N = 32$ and $N = 64$, are considered to examine whether increasing the density of question-agnostic global observations is sufficient for long-video understanding.

Although uniform sampling provides broad temporal coverage, its observation locations are independent of the question. Consequently, short-lived or fine-grained evidence may be represented by only an isolated frame or may not be observed with sufficient temporal context.

3.2 Temporal Re-observation

Temporal re-observation reallocates the visual budget from global uniform sampling to question-relevant temporal regions. The agent first uses a small set of coarse observations to identify candidate windows and then samples additional frames within those windows.

For the multi-window agent without structured memory, the total number of visual observations is expressed as

$$B = B_c + KB_l, \quad (2)$$

where B_c denotes the number of coarse global frames, K is the number of selected temporal windows, and B_l is the number of locally sampled frames within each window. This formulation allows the method to retain the same total frame budget as a uniform-sampling baseline while changing how that budget is distributed.

The benefit of temporal re-observation does not arise from observing more frames. Instead, it increases the likelihood that the available visual budget is concentrated around question-relevant events. For example, questions involving opening captions can direct observations toward the beginning of the video, whereas questions involving specific actions or objects can focus on temporal regions suggested by coarse visual evidence.

The current agent follows a bounded two-stage observation schedule. It terminates after the predefined local-frame budget is exhausted and then performs one final prediction, rather than repeatedly increasing the observation context without limit.

3.3 Structured Segment Memory

Let T denote the video duration and let $\Delta = 60$ s be the segment duration. The video is divided into

$$M = \left\lceil \frac{T}{\Delta} \right\rceil \quad (3)$$

non-overlapping segments. The segment with zero-based index s is defined as

$$S_s = [s\Delta, \min((s+1)\Delta, T)), \quad s = 0, \dots, M-1. \quad (4)$$

This definition assigns 60 seconds to each complete segment, while the final segment may be shorter when the video duration is not an exact multiple of Δ . For each segment, a small number of representative frames are sampled and processed by the frozen base model to generate a structured memory record. Each record contains an overall scene description, visible characters or entities, major actions and interactions, relevant objects, readable text, counting cues, temporal cues, and compact retrieval keywords.

These structured fields provide a concise and retrieval-oriented representation of each segment. Compared with repeatedly scanning the complete video for every question, segment-level memory enables the system to efficiently search previously summarized video content and identify potentially relevant temporal regions.

Memory records are stored independently for each video, preventing information from different videos from being mixed. The memory is constructed once during offline processing and can subsequently be reused for multiple questions associated with the same video. During query-time reasoning, the stored memory remains fixed, avoiding unstable updates caused by incorrect intermediate predictions.

3.4 Question-Aware Memory Retrieval

Given a question q and its candidate answers $\mathcal{A}$, the system searches the structured segment memory for records that are semantically relevant to both the question and the options. The retrieval stage produces a ranked set of seed segments:

$$\mathcal{S}_{\text{seed}} = \text{Retrieve}(q, \mathcal{A}, \mathcal{M}), \quad (5)$$

where $\mathcal{M}$ denotes the structured memory bank of the current video.

The retrieved records are not used as direct evidence for answer prediction. Instead, their segment indices determine where the temporal re-observation tool should inspect the original video. In this way, memory retrieval functions as temporal localization rather than text-only question answering.

3.5 Neighbor Expansion and Temporal Merging

A retrieved seed segment may be semantically related to the target event while failing to fully cover the decisive evidence because of segment-boundary effects. To improve temporal coverage, each seed segment s is expanded to include its neighboring segments:

$$\mathcal{E}_r(s) = \{j \in \mathcal{I} \mid |j - s| \leq r\}, \quad (6)$$

$$\mathcal{I} = \{0, \dots, M - 1\}.$$

where r is the expansion radius. The main setting uses $r = 2$, corresponding to two adjacent segments before and after each seed.

Expanded regions generated from different seeds may overlap. Such regions are therefore merged into continuous temporal groups:

$$\mathcal{G} = \text{Merge} \left(\bigcup_{s \in \mathcal{S}_{\text{seed}}} \mathcal{E}_r(s) \right). \quad (7)$$

Temporal merging removes redundant observations and preserves the continuous context surrounding candidate events. Local frames are subsequently sampled from the merged temporal groups and provided to the base model for final question answering.

3.6 Visual-Grounded Answer Prediction

The framework adopts a weakly decoupled organization between video indexing and answer reasoning. Structured memory compresses long-video content into segment-level records, and question-aware retrieval uses these records to determine where relevant evidence is likely to appear. The final prediction, however, is generated from the corresponding local visual observations.

This distinction is important because textual summaries may omit fine-grained visual details, including small accessories, labels, object attributes, exact entity counts, or brief action transitions. Structured memory is therefore treated as a temporal index for visual re-observation rather than as a replacement for the original visual evidence.

Given the locally re-observed frames $\mathcal{V}_{\text{local}}$, the final prediction is obtained as

$$\hat{a} = f_{\text{VLM}}(q, \mathcal{A}, \mathcal{V}_{\text{local}}, \mathcal{M}_{\text{sel}}), \quad \hat{a} \in \mathcal{A}, \quad (8)$$

where f_{VLM} denotes the frozen vision-language model and $\mathcal{M}_{\text{sel}}$ denotes the selected structured memory records. The selected memory is retained as auxiliary guidance, while the prompt explicitly instructs the model to treat the re-observed video frames as the primary evidence.

4 Experiments

4.1 Dataset and Evaluation Settings

We evaluate the proposed framework on LVBench [2], a public benchmark for long-video understanding. To conduct a computationally feasible evaluation, we construct LVBench-30, an evaluation subset containing three long videos, with ten multiple-choice questions sampled from each video, resulting in 30 question-answer pairs in total. The subset covers visible-text recognition, scene and object understanding, action and event reasoning, counting, causal reasoning, and other composite video-understanding questions. The three evaluation videos have durations of approximately 102, 42, and 35 minutes, respectively, resulting in roughly three hours of video content.

All methods use the same Qwen3-VL-8B-Instruct checkpoint without model training or fine-tuning. The compared methods differ only in how visual observations are selected and organized. Uniform U32 and U64 perform question-agnostic global sampling, the multi-window agent reallocates a fixed visual budget to question-dependent temporal windows, and the final method retrieves candidate regions from persistent structured memory before performing local temporal re-observation. Predictions are evaluated as multiple-choice labels, and accuracy is calculated over all 30 questions.

4.2 Main Results

Table 1 presents the overall comparison. Uniform U32 and U64 both achieve 33.33% accuracy, indicating that simply doubling the number of uniformly sampled frames does not improve performance on this subset. The multi-window agent reaches 43.33% under the same 32-frame visual budget as U32, showing that question-aware allocation is more effective than uniform global sampling. Structured-Memory + Expansion obtains the best result, reaching 53.33%, and improves by 20.00 percentage points over both uniform baselines.

4.3 Per-video Results

Table 2 shows that the overall gain is not produced by a single video. Structured-Memory + Expansion achieves the best result on Videos A and B, improving to 5/10 and 6/10, respectively. On Video C, the multi-window agent performs best with 6/10, while the final method obtains 5/10. This variation indicates that persistent memory is beneficial overall but does not uniformly dominate question-guided local obser-

Table 1. Main results. All methods use the same Qwen3-VL base model. U32 and U64 perform uniform sampling and direct question answering, whereas the agent variants use question-aware temporal re-observation and structured memory.

Method	Memory	Tool Use	Observation Strategy	Accuracy
Uniform U32	✗	✗	Uniform 32-frame sampling	10/30 (33.33%)
Uniform U64	✗	✗	Uniform 64-frame sampling	10/30 (33.33%)
Multi-window Agent	✗	✓	Question-guided multi-window sampling	13/30 (43.33%)
Structured-Memory + Expansion	✓	✓	Memory-guided temporal re-observation	16/30 (53.33%)

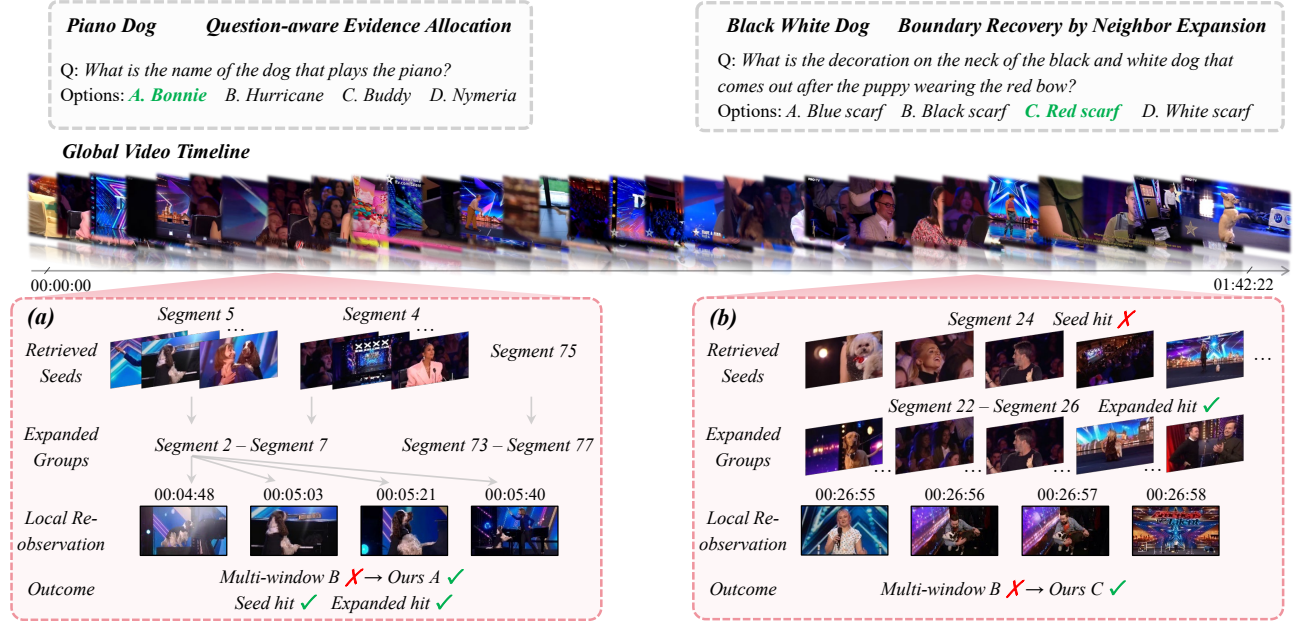


Figure 2. Temporal evidence allocation and boundary recovery. Panel (a) visualizes question-aware retrieval, temporal expansion, and local re-observation for the piano-dog question. Panel (b) shows how neighbor expansion recovers boundary-missed evidence for the black-and-white dog question.

Table 2. Per-video accuracy. Videos A, B, and C correspond to the approximately 102-, 42-, and 35-minute videos, respectively. Each video contains ten questions.

Method	A	B	C	Overall
Uniform U32	3/10	3/10	4/10	10/30
Uniform U64	4/10	3/10	3/10	10/30
Multi-window Agent	2/10	5/10	6/10	13/30
Structured-Memory + Expansion	5/10	6/10	5/10	16/30

vation on every video, particularly when coarse observations already provide useful temporal cues.

4.4 Question-type Results

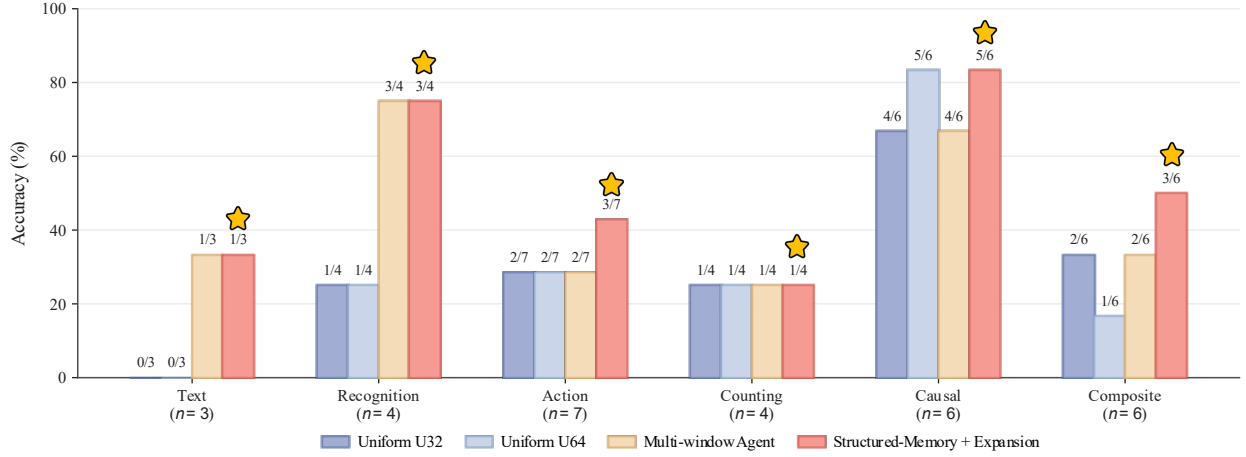
Table 3 provides a more detailed view of the gains. The clearest improvement occurs on scene and object questions, where both agent variants reach 3/4 compared with 1/4 for uniform sampling. Structured memory also improves action and event questions from 2/7 to 3/7 and other composite questions from 2/6 to 3/6. Visible-text recognition remains difficult,

with only 1/3 correct for both agent variants, while counting remains unchanged at 1/4 across all methods. For cause and reason questions, Structured-Memory + Expansion matches the strongest uniform baseline at 5/6 rather than producing an additional gain. These results support the conclusion that improved temporal localization helps several categories but does not by itself solve fine-grained OCR and counting.

Figure 3 complements Table 3 by making the category-level performance differences more explicit. The agent variants provide the largest improvement on recognition questions, where both reach 3/4 compared with 1/4 for uniform sampling. Structured-Memory + Expansion further improves action and composite questions over the multi-window agent, indicating that persistent memory is particularly useful when the answer depends on identifying a specific event or combining multiple visual cues. In contrast, all methods remain at 1/4 on counting questions, while the limited improvement on text questions con-

Table 3. Results across different question types. The best result in each category is highlighted in bold.

Question Type	Questions	Uniform U32	Uniform U64	Multi-window Agent	Structured-Memory + Expansion
Text	3	0/3	0/3	1/3	1/3
Recognition	4	1/4	1/4	3/4	3/4
Action	7	2/7	2/7	2/7	3/7
Counting	4	1/4	1/4	1/4	1/4
Causal	6	4/6	5/6	4/6	5/6
Composite	6	2/6	1/6	2/6	3/6
Total	30	10/30	10/30	13/30	16/30

**Figure 3. Accuracy across question types.** Values above the bars report the number of correct predictions over the number of questions in each category. Question types follow the category labels used in the evaluation manifest. The comparison highlights the category-specific effects of question-guided temporal observation and structured memory.**Table 4. Retrieval coverage and boundary recovery.** Expansion includes two neighboring segments on each side of every seed and merges overlapping temporal groups.

Metric	Count	Rate
Seed retrieval hit	20/30	66.67%
Expanded temporal hit	25/30	83.33%
Recovered by expansion	5/30	16.67%
Correct with seed hit	12/30	40.00%
Correct with expanded hit	14/30	46.67%
Correct without expanded hit	2/30	6.67%
Final accuracy	16/30	53.33%

firmly that temporal localization alone does not resolve fine-grained OCR. The causal category also shows that the final method does not consistently exceed the strongest uniform baseline across every question type.

4.5 Retrieval and Boundary Recovery

Table 4 analyzes the temporal localization process. The initially retrieved seed segments overlap with the annotated evidence in 20 of 30 cases. After neighbor expansion, coverage rises to 25 of 30 cases, mean-

ing that five questions are recovered specifically by adding adjacent temporal context. Among the final correct predictions, 12 have a seed hit and 14 have an expanded hit. The remaining gap between the 83.33% expanded hit rate and the 53.33% final accuracy shows that temporal localization is necessary but not sufficient: the model may inspect the correct interval yet still misread visible text, miscount entities, or misunderstand a short action sequence.

Two final answers are correct even though the expanded windows do not overlap the annotated evidence interval. These cases may result from broader contextual cues, incomplete temporal annotations, or correct predictions made without localized evidence. They should therefore not be interpreted as successful temporal grounding.

Figure 2 illustrates how the final agent converts full-video memory into localized visual evidence. In the piano-dog case, the ranked seeds are S_5 , S_4 , and S_{75} . Segments S_4 and S_5 overlap the annotated interval from 00:04:45 to 00:05:49, whereas S_{75} is a semantically similar dog-performance distractor. Neighbor expansion merges the first two seeds into

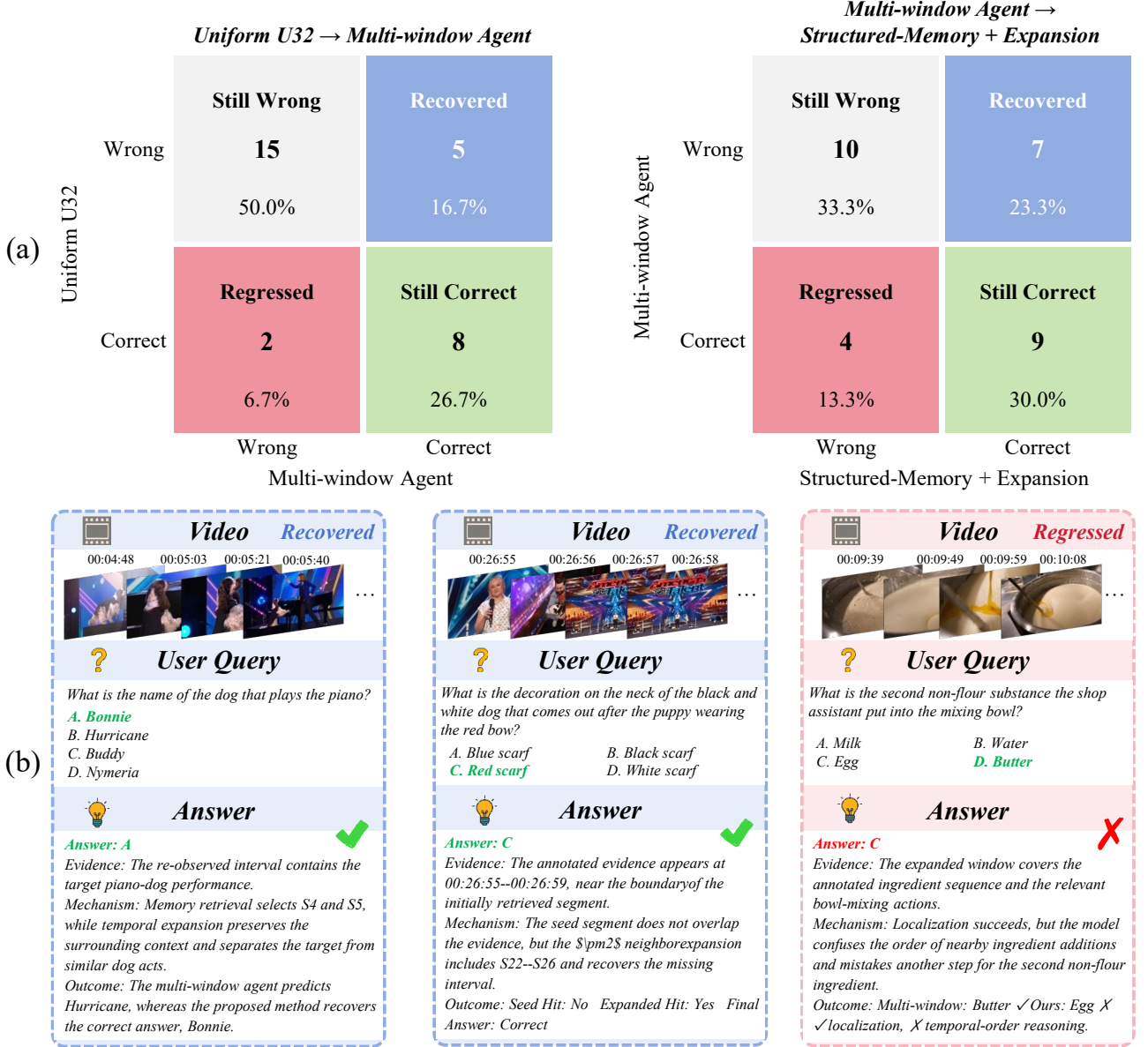


Figure 4. Correctness transitions and representative qualitative cases. Panel (a) reports question-level correctness transitions from Uniform U32 to the multi-window agent and from the multi-window agent to Structured-Memory + Expansion. Panel (b) presents two recovered cases and one regression case. The first case is recovered through structured-memory retrieval, the second demonstrates boundary evidence recovered by neighbor expansion, and the third shows that successful temporal localization does not necessarily guarantee correct temporal-order reasoning.

S_2 – S_7 and expands the distractor to S_{73} – S_{77} . Local re-observation within the retrieved region then captures the target performance and changes the multi-window prediction from Hurricane to the correct answer, Bonnie. This case demonstrates that structured memory can concentrate observation around a relevant event while retaining enough temporal context to distinguish it from repeated, visually similar acts.

The black-and-white dog case isolates the contribution of neighbor expansion. The initial seed S_{24} is semantically related to the query but does not overlap the annotated evidence at 00:26:55–00:26:59, which

lies in S_{26} . Expanding the seed by two neighboring segments on each side produces S_{22} – S_{26} , changing the retrieval status from a seed miss to an expanded hit. The subsequent local observations expose the neck decoration and support the correct answer, Red scarf. This example provides a concrete explanation for the five boundary cases recovered by expansion in Table 4.

Together, the two cases separate the roles of the three stages: structured memory supplies coarse semantic localization, neighbor expansion compensates for segment-level temporal error, and local re-

observation returns to the original video for fine-grained answer evidence. Unlike Figure 4, which analyzes how predictions change across methods, Figure 2 focuses specifically on where the system observes and how the final evidence windows are formed.

4.6 Correctness Transition and Qualitative Analysis

Figure 4 provides a more detailed view of how the performance gains in Table 1 are obtained. When moving from Uniform U32 to the multi-window agent, five previously incorrect questions are recovered, while two previously correct questions become incorrect. Eight questions remain correct and fifteen remain incorrect, resulting in a net gain of three correct answers. The transition from the multi-window agent to Structured-Memory + Expansion recovers seven additional questions but introduces four regressions. Nine questions remain correct and ten remain incorrect, again yielding a net gain of three. These transitions show that the improvement is not produced by uniformly preserving all previous successes; instead, the stronger observation strategy solves a new subset of questions while occasionally changing previously correct predictions.

The qualitative cases explain the mechanisms behind these transitions. In the first recovered case, the multi-window agent confuses the target piano-dog performance with another semantically similar dog act. Structured memory retrieves segments S_4 and S_5 , and temporal expansion preserves the surrounding visual context, allowing the final visual re-observation to recover the correct answer, Bonnie. This example illustrates how persistent segment memory can disambiguate repeated events that are visually and semantically similar.

The second recovered case isolates the contribution of neighbor expansion. The initially retrieved seed segment does not overlap the annotated evidence at 00:26:55–00:26:59. After applying a ± 2 segment expansion, the resulting interval S_{22} – S_{26} includes the missing boundary evidence, enabling the system to identify the red scarf correctly. This case directly supports the improvement from a 66.67% seed hit rate to an 83.33% expanded hit rate reported in Table 4.

The regression case reveals a remaining limitation. Although the expanded window overlaps the annotated bowl-mixing sequence, the final model confuses the order of nearby ingredient additions and predicts Egg instead of Butter. The multi-window agent answers this question correctly, whereas Structured-

Memory + Expansion does not. This example indicates that improved temporal localization addresses where to observe, but does not fully resolve how to interpret fine-grained action order within the localized interval.

4.7 Error Analysis

The remaining errors of the final system can be broadly categorized into visible-text recognition, counting, action and event reasoning, retrieval ambiguity, and local reasoning errors.

OCR and visible text. Some questions require recognizing names, captions, signs, or text printed on small objects. Even when the correct temporal window is retrieved, the model may fail because the relevant text is small, partially occluded, blurred, or visible for only a few frames. Local re-observation improves the probability of observing the text but does not fully resolve fine-grained OCR errors.

Counting. Questions involving the number of animals, objects, food items, or repeated actions remain challenging. Uniform sampling may omit individual instances, whereas locally sampled frames may still be too sparse to support reliable counting. Frame-level aggregation or an adaptive counting verifier could provide more stable estimates.

Action and event reasoning. Questions containing temporal relations such as *before*, *after*, and *then* require continuous tracking of event order. Although structured memory can identify a relevant temporal region, the locally re-observed frames may not cover the entire event sequence with sufficient temporal density. Longer continuous observation or hierarchical event tracking may be required for such cases.

Retrieval ambiguity. Some errors occur when the memory retrieval stage selects a semantically related but incorrect segment. For example, a question about one dog performance may retrieve another dog performance from the same talent-show video. In this case, the retrieved segment is topically relevant but does not contain the exact evidence required by the question.

Final reasoning error. In several cases, the expanded temporal group overlaps with the annotated evidence, but the final answer remains incorrect. This indicates that temporal localization and evidence coverage are necessary but not sufficient. The base model must still accurately interpret fine-grained objects, visible text, entity counts, and temporal relations within the localized visual input.

Overall, the results reveal a clear shift in the primary bottleneck. Structured memory and neighbor expansion substantially improve where the system

looks, while the remaining errors are increasingly determined by how precisely the localized evidence is interpreted.

4.8 Limitations and Future Directions

The current evaluation is limited to a 30-question subset of LVBench, so the reported results should be viewed as an initial validation rather than a complete benchmark study. Although the selected videos cover different durations and question types, experiments on a larger portion of LVBench are still needed to verify the robustness of the method across broader video domains. Another limitation lies in the structured memory itself, since the memory records are generated by the frozen base model and may contain incomplete descriptions or lose fine-grained visual details during compression. This issue is especially noticeable in questions involving small text, subtle object attributes, counting, or short actions. In addition, the current temporal observation strategy uses fixed sampling settings, which may not be suitable for all question types. Future work will therefore focus on more adaptive local observation, allowing the system to increase temporal density when the available evidence is still insufficient. The final answer should also continue to rely on the original visual observations rather than textual memory alone, because memory summaries are useful for locating relevant regions but cannot fully preserve all details required for the final decision.

5 Conclusion

This report presents a training-free long-video QA agent based on structured memory and temporal re-observation. The LVBench-30 evaluation shows that increasing uniform frames from 32 to 64 does not improve accuracy, while question-aware local re-observation and structured memory produce clear gains. The final Structured Memory + Neighbor Expansion system improves accuracy from 33.33% to 53.33% and raises temporal evidence coverage to 83.33%. The study suggests that long-video understanding benefits from careful orchestration of memory, retrieval, and local perception, rather than from uniformly increasing sampled frames alone.

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